

Mid Term Test
IT509 Data Mining and Data Warehousing
September, 2018
Marks: 40 Time: 1 hour 15 minutes

1. State true or false: 1.5×5=7.5
- a) Single linked distance is a dissimilarity metric used in hierarchical clustering.
 - b) The complete linkage-hierarchical algorithm finds compact but globular clusters.
 - c) K-means is extremely sensitive to cluster center initializations.
 - d) A data warehouse stores current data.
 - e) Outlier removal is the most appropriate strategy for data cleaning before performing clustering analysis.
2. Fill up the blanks in the following sentences. 1.5×5=7.5
- (a) The _____ clustering algorithm finds nested clusters.
 - (b) Termination condition of k-means clustering is _____.
 - (c) One of the clustering algorithms that can find arbitrarily shaped clusters is _____.
 - (d) Computational complexity of a hierarchical clustering algorithm is _____.
 - (e) A proximity measure means either a similarity measure or a _____ measure.
3. (a) What is clustering? 2
- (b) State its importance. 2
- (c) Categorize different clustering algorithms. 2
- (d) State some important requirements that a clustering algorithm need to fulfil. 4
4. Discuss about hierarchical clustering algorithms. 5
5. a) What are the advantages and disadvantages of the k-means algorithm? 3
- b) What are the advantages and disadvantages of the PAM algorithm? 3
- c) How does CLARA algorithm works. 4